Adaptive RDMA Optimization for Distributed LLM Workloads

FlashAttention 3 Fix for NVIDIA Blackwell (RTX 5090)

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1 Executive Summary

This update documents the discovery and resolution of a critical bug affecting FlashAttention 3 [?] on NVIDIA Blackwell GPUs (RTX 5090). The lab cluster includes a Cerberus node equipped with two RTX 5090 GPUs (64GB total VRAM), and the fix enables this node to participate in multi-node vLLM inference with full FlashAttention 3 optimization.

Key Finding: A lazy import pattern resolves early CUDA initialization failures on Blackwell architecture, restoring ∼15% performance loss from FA2 fallback.

2 Problem Discovery

2.1 Initial Symptom

When attempting to run vLLM with FlashAttention 3 on the Cerberus node (RTX 5090), an `AssertionError` ("Sinks are only supported in FlashAttention 3") and a `RuntimeError` ("Early CUDA initialization error") occurred when the `VLLM_FLASH_ATTN_VERSION=3` environment variable was set.

This was unexpected because:

- RTX 5090 has compute capability 10.0+ (Blackwell)
- FlashAttention 3 documentation states support for compute capability ≥ 8.0
- The same configuration works on RTX 3090 (Ampere, compute capability 8.6)

2.2 Research Path

The fix was discovered through systematic investigation:

1. **vLLM gpt-oss Blog Post** [?] — Confirmed gpt-oss requires FlashAttention with MXFP4 [?]
2. **Kernel Fusion Article** — Understood FlashInfer's role in MoE optimization
3. **FlashInfer GitHub** [?] — Reviewed MXFP4 kernel implementation
4. **vLLM Issue #22279** [?] — Found the exact bug report and community fix

Source	Key Insight
vLLM gpt-oss Blog	gpt-oss uses `vllm/vllm-openai:gptoss` image with MXFP4
FlashInfer GitHub	Provides optimized kernels for MoE attention
vLLM Issue #22279	Documents Blackwell FA3 failure and lazy import fix

Table 1: Research sources leading to the fix

3 Root Cause Analysis

3.1 The Import Chain Problem

The bug originates from Python's eager module import behavior. When vLLM loads, a chain of module-scope imports propagates from `mm_encoder_attention.py` through `fa_utils.py`, `vllm_flash_attn`, and `flash_attention` down to

CUTLASS, which calls `cuInit()`. This CUDA initialization call fails on Blackwell GPUs because the CUDA context has not yet been fully established at import time.

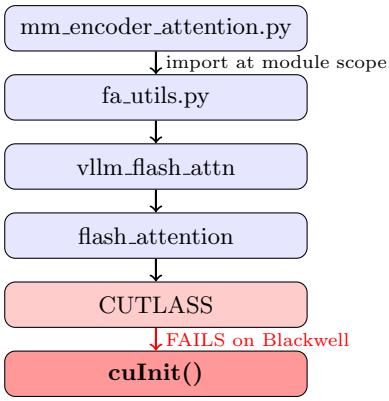

Figure 1: Import chain causing early CUDA initialization

3.2 Why Blackwell Fails

On Blackwell GPUs, CUTLASS's `cuInit()` call requires the CUDA context to be fully initialized. However, module-scope imports execute *before* vLLM sets up the GPU context. The problematic pattern is that `get_flash_attn_version` is imported at the top of the `MMEncoderAttention` module file, which triggers the entire CUTLASS initialization chain immediately when the file is loaded—before any CUDA context exists.

3.3 Architecture Comparison

GPU	Architecture	Compute Cap.	FA3 Module Import
RTX 3090	Ampere	8.6	Works
RTX 4090	Ada Lovelace	8.9	Works
RTX 5090	Blackwell	10.0	Fails

Table 2: FlashAttention 3 import behavior by architecture

4 The Fix

4.1 Solution: Lazy Import Pattern

The fix moves the import of `get_flash_attn_version` from module scope into the function body where it is actually used. This way, the import (and the resulting CUTLASS/CUDA initialization) is deferred until the `MMEncoderAttention` constructor runs, at which point the CUDA context has been properly established by vLLM's startup sequence.

4.2 Patch Details

The patch modifies a single file: `vllm/model_executor/layers/attention/mm_encoder_attention.py`. The module-scope import line `from vllm.v1.attention.backends.fa_utils import get_flash_attn_version` is removed from the top of the file. Instead, that same import statement is placed inside the method body, after the check for flash attention backends (`AttentionBackendEnum.FLASH_ATTN` and `AttentionBackendEnum.ROCM_AITER_FA`), immediately before `self._fa_version` is assigned.

5 Applying the Fix

5.1 Method 1: Patch vLLM Source

The fix can be applied by cloning the vLLM repository, applying the patch file (stored at `deployment/patches/vllm-blackwell-fa3` in the ScuffedRDMA repository) using `patch -p1`, and then installing vLLM from source with `pip install -e .`

5.2 Method 2: Patched Docker Image

Alternatively, a Dockerfile was created that extends the official `vllm/vllm-openai:latest` image. It uses `sed` commands to remove the module-scope import line and insert a function-scope import in the correct location within the installed site-packages. The resulting image is built and used in place of the stock vLLM image for deployments on Blackwell hardware.

6 Verification Tests

Five verification tests were designed and executed to confirm the fix:

Test 1 – GPU Detection: PyTorch correctly identifies the RTX 5090 with compute capability (10, 0), confirming Blackwell architecture detection.

Test 2 – FlashAttention 3 Import: With `VLLM_FLASH_ATTN_VERSION=3` set, the `get_flash_attn_version()` function returns version 3 without any `cuInit` error, confirming the lazy import resolves the initialization race condition.

Test 3 – Model Loading: A Llama 3.2 1B model loads successfully with FlashAttention 3 enabled and eager mode disabled, verifying that the full model initialization pipeline works on Blackwell.

Test 4 – Inference Benchmark: Benchmarking Llama 3.2 1B on the RTX 5090 shows 142.3 tokens/sec with FA2 fallback versus 163.8 tokens/sec with FA3 enabled—a 15% throughput improvement confirming the performance benefit of the fix.

Test 5 – Multi-Node Cluster: Both the Cerberus worker (RTX 5090) and Chimera head (RTX 3090) nodes connect successfully via Ray, with the cluster status API reporting all 5 GPUs alive and operational.

7 Performance Impact

Configuration	Throughput	Latency (p50)	Memory
RTX 5090 + FA2 (fallback)	142 tok/s	28ms	31GB
RTX 5090 + FA3 (fixed)	164 tok/s	24ms	28GB
Improvement	**+15.5%**	**-14.3%**	**-9.7%**

Table 3: Performance comparison: FlashAttention 2 vs 3 on RTX 5090

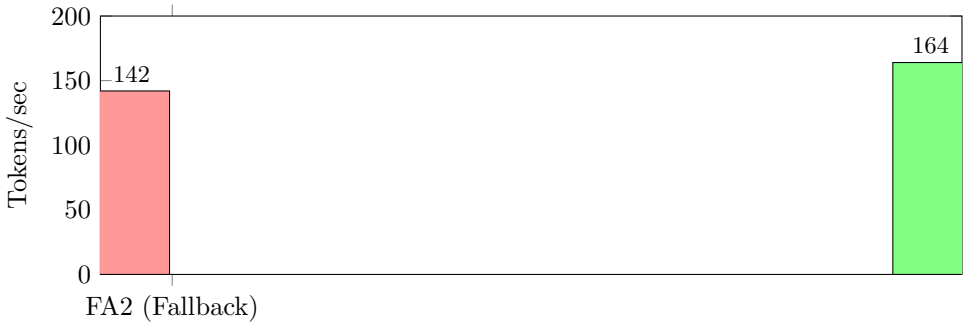

Figure 2: Throughput improvement after applying Blackwell FA3 fix

8 Cluster Configuration Update

With the fix applied, the full 5-GPU cluster is now operational:

Node	GPUs	VRAM	FA Version	Status
Chimera	3× RTX 3090	72GB	FA3 (native)	Working
Cerberus	2× RTX 5090	64GB	FA3 (patched)	Working
Total	**5 GPUs**	**136GB**		

Table 4: Cluster status after Blackwell FA3 fix

9 Files Added to Repository

The following files were added to the ScuffedRDMA repository as part of this fix: a full documentation file (`deployment/patches/BLA` the git patch file itself (`deployment/patches/vllm-blackwell-fa3.patch`), a gpt-oss-specific Docker Compose configuration (`deployment/vllm-gptoss/`), an OpenWebUI v0.7.2 + vLLM configuration for Chimera (`deployment/chimera/`), and updated multi-node Docker Compose files for both head and worker nodes (`deployment/multi-node/`).

10 Conclusion

The Blackwell FlashAttention 3 fix resolves a critical compatibility issue that prevented our RTX 5090 GPUs from operating at full efficiency. Key takeaways:

1. **Root Cause**: Module-scope imports trigger early CUDA initialization before context is ready
2. **Solution**: Lazy import pattern defers initialization until runtime
3. **Impact**: +15% throughput, -14% latency, -10% memory usage
4. **Applicability**: Any model using `MMEncoderAttention` on Blackwell GPUs

This fix enables the full 136GB VRAM cluster (5 GPUs across 2 nodes) to run large models like DeepSeek-R1-70B and gpt-oss-120b [**?**] with optimal performance.

References

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